Fiscal Year:	FY 2006	**Task Last Updated:**	FY 03/07/2006
PI Name:	Yamauchi, Mitsuo Ph.D., D.D.S.		
Project Title:	Mechanisms of Bone Loss Induced by Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	27599	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	11/01/2002	**End Date:**	10/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mochida, Yoshiyuki Ph.D. (University of North Carolina)		
Grant/Contract No.:	NAG2-1596		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The long-term goal of this research is to understand, on the molecular, cellular and biochemical levels, the mechanisms of bone loss caused by the condition of microgravity. Our underlying hypothesis is that cells in weight-bearing bones, when subjected to microgravity/skeletal unloading, synthesize and deposit post-translationally altered type I collagen that does not mineralize properly leading to bone loss. During the last grant period, by employing a rat tail suspension model, we found that skeletal unloading resulted in accumulation of collagen with elevated lysine hydroxylation and altered cross-linking pattern in weight-bearing bones and that the alterations were bone-region specific. In addition, the lysyl hydroxylase gene expression was significantly up-regulated in these bones. In this renewal, as a natural extension of these findings, we now propose two major objectives: 1. to further define the skeletal unloading-induced collagen alterations in weight-bearing bones by employing a rat-tail suspension model, and 2. to investigate whether or not		

Task Description:	elevated lysine hydroxylation of collagen is a cause of impaired collagen and poor mineralization in vitro. The first aim will be achieved by biochemical analysis, newly developed bioimaging and molecular technologies that are complementary to one another. The second aim will be achieved by evaluating the effects of overexpression of lysyl hydroxylase genes in osteoblastic cell line on a type I collagen molecule, cross-linking and mineralization in vitro. These extensive and complementary studies will hopefully provide insights into the mechanisms of bone loss occurred during spaceflight and would help develop its preventive strategy. To understand the mechanisms of bone loss/defects induced by skeletal unloading/microgravity. This study may help to develop strategies to prevent the bone loss that occurs during spaceflight/unloading.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Recently, we successfully generated osteoblast-derived clones that overexpress or underexpress lysyl hydroxylase 2b (LH2b), an enzyme important for lysine (Lys) hydroxylation in the collagen telopeptides/cross-linking, and partially characterized the effects. The data clearly demonstrated a critical role of LH2b in determining collagen cross-linking pathway (Pornprasertsuk et al, J Bone Min Res 2004) and in regulating collagen fibril growth and subsequent mineralization (Pornprasertsuk et al, J Bone Min Res 2005). In order to further understand the potential roles of Lys hydroxylation of collagen in mineralization, we also generated osteoblastic cell-derived clones that overexpress LH1 and 3 by employing the same strategy. After several clones were established, we performed an in vitro mineralization assay. The results demonstrated that higher levels of LH1 and LH3 isoforms exerted differential effects on in vitro mineralization. The former slightly delayed the mineralized nodule formation while the latter promoted the process in comparison with controls. These results together with our recent study on LH2b suggest that the three LH isoforms have distinct roles in collagen modifications leading to differential effects on collagen matrix mineralization (Pornprasertsuk et al, Proc Int Conf Chem Biol Min Tis, 2005). We have been further characterizing the clones overexpressing LH3 to understand the potential mechanisms of the accelerated mineralization. Lysyl oxidase (LOX), another collagen modifying enzyme, is critical to initiate the process of collagen cross-linking. Recently, several LOX-like proteins (LOXL1-4) have been identified and partially characterized. Since a major objective of this grant is to understand the roles of collagen modifications in bone physiology, we decided to investigate if those LOX isoforms are expressed in osteoblastic cells. The mRNA expression of LOX and LOXL1-4 in MC3T3-E1 cells were screened by RT-PCR and quantitatively analyzed by real-time PCR. The results demonstrated that LOX and all LOXLs, except LOXL2, were expressed in this cell line and that the expression pattern during the course of cell differentiation/matrix mineralization was distinct from one another. This indicates that the expression of LOX and its isoforms are highly regulated in a differentiation-stage specific manner. These newly identified LOX isoforms in osteoblastic cells may have distinct roles in collagen matrix stabilization and subsequent mineralization (Atsawasuwan et al, Biochem Biophys Res Commun 2005). All of these studies indicate that the post-translational modifications of collagen (Lys hydroxylation/cross-linking) are important to regulate the process of collagen fibril maturation and subsequent mineralization in bone. Furthermore, these important modifications are likely orchestrated by a number of isoforms of LH and LOX expressed at specific stages of osteoblast differentiation/maturation. Thus, bone loss caused by microgravity may be due in part to altered expression pattern of multiple LH and LOX genes in osteoblasts in response to unloading. Partially supported by this grant, we also started investigating the roles of biglycan, a member of small leucine-rich proteoglycan family, in osteoblast differentiation. We found that when biglycan was over- or under-expressed, osteoblast differentiation/mineralization was accelerated or delayed (Parisuthiman et al J Bone Min Res 2005). Our further study suggested that this proteoglycan facilitates bone morphogenetic protein (BMP)-2 induced osteoblast differentiation possibly through its direct interaction with BMP-2 and its type I receptor, BMPRIB/ALK-6 (Mochida et al, Adv Exp Bio Med, in press).
Bibliography Type:	Description: (Last Updated: 08/11/2020)
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